Tutorial - Group Theory

Solution

1. A group $(A, *)$ is called abelian if $*$ is a commutative operation

$$(g \circ h)^2 = (g \circ h) \circ (g \circ h)$$
$$= g \circ h \circ g \circ h.$$
$$= g \circ g \circ h \circ h \quad (\because h \circ g = g \circ h)$$
$$= g^2 \circ h^2$$

Ans C

2. $\langle A_{n \times n}, * \rangle$ Set of Non Singular Matrix is Non Commutative group as

✓ product of two Non Singular matrices is Non Singular matrix, Thus it is closed under binary Composition

✓ Matrix multiplication is always associative.

✓ I_n, the identity element is Identity Matrix

✓ Inverse of each Non Singular matrix exists.

✓ Matrix multiplication is Non Commutative i.e $A \times B \neq B \times A$

Thus (D) is Answer.

(3) c is false as zero does not have multiplicative inverse.

(4) $$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a\cos\theta & -b\sin\theta \\ a\sin\theta & b\cos\theta \end{bmatrix} = \begin{bmatrix} a\cos\theta & -a\sin\theta \\ -a\sin\theta & b\cos\theta \end{bmatrix}$$

$$\Rightarrow -b\sin\theta = -a\sin\theta$$

$$\Rightarrow \sin\theta(a-b) = 0$$

$$\Rightarrow a = b \ \& \ \sin\theta = 0$$

$$\theta = n\pi, \ n = \text{integer}$$

<u>Ans</u> A.

(5) $$x * y = \bar{x} + y$$

$$z * x = (x * y) * x$$

$$= (\bar{x} + y) * x$$

$$= \overline{\bar{x} + y} + x$$

$$\Rightarrow x\bar{y} + x \Rightarrow x[1 + \bar{y}] = x$$

(B) is the correct option

(6) (D) None of the above as only binary composition closure and associative property hold. Thus it's a Semigroup.

(7) (B) Set of Non Singular matrix is a group

(A) is false as $(-1)\times(-1) = +ve$. violates closure.

(C) is false As Non Singular matrix has inverse but all $n\times n$ matrices are not inverse Thus inverse of each matrix does not exist.

(8) A : Does not form group as inverse of each string does not exist.

(a) $S_1 \in Z^*$, $S_2 \in Z^* \Rightarrow S_1 S_2 \in Z^*$.

$(S_1 * S_2) * S_3 = S_1 * (S_2 * S_3)$

$(S_1 S_2) * S_3 = S_1 * (S_2 S_3)$

$S_1 S_2 S_3 = S_1 S_2 S_3$

Thus associative

(c) λ – Null string is identity element

$$S_1 * \lambda = \lambda * S_1 = S_1$$

$$S_1 = S_1 = S_1$$

(d) $\boxed{S_1 * S_2 = \lambda = S_2 * S_1}$

If we Concatenate two strings then we Cannot get NULL strings. Thus inverse does not exist.

9. using $(a \times x) + (a \times y) = c$ (1)

Case 1

$$a + c = c$$

Case 2 $b + c = c$

Case 3 $c + b = c$

using Case 1

$a \times x = a$ and $a \times y = c$

$\Rightarrow x = a$, and $y = c$

one pair (a, c)

using Case 2

$a \times x = b$ and $a \times y = c$

$x = b$ and $y = c$

(b, c) is 2nd pair

Case 3 (c, b) is 3rd pair

9. Cont...

Now we will take equation 2

$$(b \times x) + (c \times y) = c$$

To check how many pairs (x, y) satisfy equation 2.

pair ① $b \times a + c \times c = c$

$x = a,\ y = c$

$\boxed{b + b = c}$ false.

Pair 2

$x = b,\ y = c$

$(b \times b) + (c + c) = c$

$c + b = c$ = True

Pair 3

$x = c,\ y = b$

$(b \times c) + (c \times b) = c$

$\boxed{a + c = c}$ True

Thus 2 pairs (b, c) and (c, b) satisfy

Ans C.

Q10 $ABCD = I$

$$A^{-1}ABCDD^{-1} = A^{-1}ID^{-1}$$

multiplying left hand side by A^{-1} and Right hand side by D^{-1}

$$IBCI = A^{-1}D^{-1}$$

$$BCC^{-1} = A^{-1}D^{-1}C^{-1}$$

$$B = A^{-1}D^{-1}C^{-1}$$

Thus $B^{-1} = (A^{-1}D^{-1}C^{-1})^{-1}$

$$= (C^{-1})^{-1}(D^{-1})^{-1}(A^{-1})^{-1}$$

$$\boxed{B^{-1} = CDA}$$

Ans B.

(11) C $4 \times_{15} 4 = 1$ or $4 \times 4 \equiv 1 \bmod 15$

and $7 \times_{15} 13 = 1$ or $7 \times 13 \equiv 1 \bmod 15$

$4^{-1} = 4$ and $7^{-1} = 13$

(12)

(12) ✓ product of two upper triangular Non Singular matrix is also a upper triangular matrix.

✓ Matrix multiplication is always associative

✓ $\begin{bmatrix} 1 & 0 & 0 \\ 8 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is also an upper triangular matrix which is an identity element

✓ $\begin{bmatrix} a & f & e \\ 0 & b & d \\ 0 & 0 & c \end{bmatrix}^{-1} = \frac{1}{abc}\begin{bmatrix} bc & -fc & fd-be \\ 0 & ac & -ad \\ 0 & 0 & ab \end{bmatrix}$

which is again a upper triangular matrix with determinant Nonzero. Thus it is a group.

(13) $E[p^2q]$ = Euler function : Number of +ve integers less than N and Co-prime to N.

$E[N] = N \times \left(1-\frac{1}{p_1}\right)\left(1-\frac{1}{p_2}\right)\cdots\left(1-\frac{1}{p_n}\right)$

$N = p_1^a p_2^b p_3^c$ here $p_1 = p$ & $p_2 = q$

$E[p^2q] = p^2q\left[1-\frac{1}{p}\right]\left[1-\frac{1}{q}\right] = p(p-1)(q-1)$

Ans D

14 Inverse exist only if Number is Coprime to m in modulo m.

(3,10) are Coprime

$\therefore$ 3^{-1} exist as $3^{-1} = 7$

$3 \times_{10} 7 = 1$ Thus (c) is false statement.

(15) $\underbrace{x * x = e}$ $\Rightarrow x^{-1} * x * x = x^{-1} * e$

$\Downarrow$

$\boxed{x = x^{-1}}$ $\Rightarrow (x^2)^k = e^k = e$

Similarly $\boxed{y = y^{-1}}$ $\Rightarrow (y^2)^k = (e)^k = e.$

Alt

$x * y * x * y = e$

$\underbrace{x^{-1} * x}_{e} * y * x * \underbrace{y * y^{-1}}_{e} = x^{-1} * e * y^{-1}$

$\Rightarrow y * x = x^{-1} * y^{-1}$

$(\because x^{-1} = x,\ y^{-1} = y)$

$\Rightarrow \boxed{y * x = x * y}$ Commutative for x & y.

$*$

Case 1

$x^{even} * y^{even}$

$\Rightarrow e * e = e$

Case 2

$x^{even} * y^{odd} = y.$

Case 3

$x^{odd} \cdot y^{even} = x$

Case $x^{odd} * y^{odd} = x * y.$

each element $a \in G$ has form $x^{k_1} y^{k_2}$ for some k_1 & $k_2 \in I$

k_1 = even/odd , k_2 = even/odd.

Total 4 elements possible in G (*)